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United States Patent Application Publication

20250271698

Kind Code

A1

Publication Date

August 28, 2025

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DISPLAY APPARATUS

Abstract

A display apparatus includes an optical member with translucency, a transmissive member, an electro-optical member, and a housing. The optical member includes a front surface corresponding to an observation side and a back surface lying behind the front surface. The transmissive member is provided on a peripheral edge of the back surface of the optical member. The transmissive member has a transmittance of 5% or more. The electro-optical member is provided on a side of the back surface of the optical member. The electro-optical member has optical characteristics varying with applied voltage. The housing includes a protruding portion. The protruding portion covers an end portion of the electro-optical member and protrudes toward the back surface of the optical member. An end portion of the transmissive member is located between an end portion of the optical member and the protruding portion.

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Appl. No.: 19/025582

Filed: January 16, 2025

Foreign Application Priority Data

JP	2024-025076	Feb. 22, 2024
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Publication Classification

Int. Cl.: G02F1/1335 (20060101); G02F1/13363 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-025076, filed on Feb. 22, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates generally to a display apparatus.

BACKGROUND

[0003] A display apparatus such as an electronic mirror has a display region inside a frame region on a front surface.

[0004] In a display mode, the display apparatus displays an image in the display region on the front surface and, in the mirror mode, causes the display region on the front surface to function as a mirror (See, for example, a patent literature JP 4418483 B2).

[0005] From the viewpoint of appearance of the display apparatus, it is desired to reduce the contour existing at the boundary between the frame region and the display region on the front surface.

SUMMARY

[0006] A display apparatus according to one aspect of the present disclosure includes an optical member with translucency, a transmissive member, an electro-optical member, and a housing. The optical member includes a front surface corresponding to an observation side and a back surface lying behind the front surface. The transmissive member is provided on a peripheral edge of the back surface of the optical member. The transmissive member has a transmittance of 5% or more. The electro-optical member is provided on a side of the back surface of the optical member. The electro-optical member has optical characteristics varying with applied voltage. The housing includes a protruding portion. The protruding portion covers an end portion of the electro-optical member and protrudes toward the back surface of the optical member. An end portion of the transmissive member is located between an end portion of the optical member and the protruding portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a schematic diagram illustrating an example of a vehicle on which an electronic mirror system including a display apparatus according to an embodiment is installed;

[0008] FIG. 2 is a front view illustrating an example of a configuration of a display apparatus according to an embodiment;

[0009] FIG. 3 is a cross-sectional view illustrating an example of a configuration of a display apparatus according to an embodiment;

[0010] FIG. 4 is a cross-sectional view illustrating an example of a configuration of a deposited layer according to an embodiment;

[0011] FIG. 5 is a table for describing transmittance of a deposited layer according to an embodiment;

[0012] FIG. 6 is a table illustrating an example of a relationship between transmittance and appearance of a deposited layer according to an embodiment;

[0013] FIG. 7 is a cross-sectional view illustrating an example of a configuration of a display apparatus according to a first modification; and

[0014] FIG. 8 is a cross-sectional view illustrating an example of a configuration of a display apparatus according to a second modification.

DETAILED DESCRIPTION

[0015] Hereinafter, an embodiment of a display apparatus according to the present disclosure will be described with reference to the drawings.

Embodiment

[0016] The display apparatus according to the embodiment is, for example, an electronic mirror. The display apparatus according to the embodiment has a frame region at an edge portion on a surface and a display region on the surface, but is devised for reducing the width of the contour existing at the boundary between the frame region and the display region.

[0017] For example, an electronic mirror system 3 can be installed in a vehicle 5 as illustrated in FIG. 1. FIG. 1 is a schematic diagram illustrating an example of the vehicle 5 on which the electronic mirror system 3 according to the embodiment is installed. The electronic mirror system 3 includes an imaging apparatus 1 and a display apparatus 2.

[0018] The imaging apparatus 1 is an in-vehicle camera installed in the vehicle 5. The imaging apparatus 1 is installed outside or inside a vehicle body 6. The imaging apparatus 1 may be installed on the rear side of the vehicle body 6 to image the rear side of the vehicle 5. The imaging apparatus 1 may be installed near the door of the vehicle body 6 to image the side of the vehicle 5. The imaging apparatus 1 may be installed on the front side of the vehicle body 6 to image the front side of the vehicle 5.

[0019] The display apparatus 2 is disposed in a vehicle interior 7. The display apparatus 2 is, for example, an electronic mirror. The display apparatus 2 has a front surface 2a and can display an image acquired by the imaging apparatus 1 on the front surface 2a. The display apparatus 2 is configured to be switchable between a display mode and a mirror mode. The display mode is a mode in which the display apparatus 2 functions as a display that displays an image captured by the imaging apparatus 1. The mirror mode is a mode in which the display apparatus 2 functions as a mirror.

[0020] In a case where the display apparatus 2 is an electronic mirror for rear visual recognition, the display apparatus 2 may be installed in the form of a rearview mirror. In this case, the front surface 2a may face the vehicle interior 7, and the shape of the front surface 2a may be the shape of a mirror surface of the rearview mirror. In a case where the display apparatus 2 is an electronic mirror for side visual recognition, the display apparatus 2 may be installed in the form of a door mirror (for example, door mirror 61). In this case, the front surface 2a may face the rear side of the vehicle body 6, and the shape of the front surface 2a may be the shape of a mirror surface of the door mirror.

[0021] In a case where the display apparatus 2 is an electronic mirror for front visual recognition, the display apparatus 2 may be installed in the form of an in-vehicle display device (for example, display device 71). In this case, the front surface 2a may face the vehicle interior 7, and the shape of the front surface 2a may be the shape of the display unit of the display device.

[0022] In FIG. 1, the imaging apparatus 1 is installed at an end portion 6a on the rear side of the vehicle body 6, and the display apparatus 2 is applied to an electronic mirror for rear visual recognition. The electronic mirror for rear visual recognition is also called an electronic rearview mirror. The imaging apparatus 1 acquires an image behind the vehicle body. The display apparatus 2 can display an image of the rear side of the vehicle body captured by the imaging apparatus 1.

[0023] The display apparatus 2 is as illustrated in FIG. 2 when viewed from the front surface 2a side. FIG. 2 is a front view illustrating an example of a configuration of the display apparatus 2 according to the embodiment. Hereinafter, a direction perpendicular to the front surface 2a is defined as a Z direction, and two directions orthogonal to each other in the front surface 2a are

defined as an X direction and a Y direction.

[0024] The front surface **2a** has, for example, a rectangular shape with rounded corners in the XY plane view. The front surface **2a** has a frame region **2a1** and a display region **2a2**. The frame region **2a1** is a portion serving as an edge on the front surface **2a**. The display region **2a2** is a portion inside the frame region **2a1** on the front surface **2a**.

[0025] In the display mode, the display apparatus **2** displays an image in the display region **2a2** on the front surface **2a**. In the mirror mode, the display apparatus **2** causes the display region **2a2** on the front surface **2a** to function as a mirror in the mirror mode.

[0026] The display apparatus **2** may have a function of a variable reflectance mirror (VRM). In a case where the display apparatus **2** has the VRM function, the reflectance of the display region **2a2** may be varied with the illuminance or the like in the vicinity of the front surface **2a** in the mirror mode. For example, in the mirror mode, in a case where the illuminance near the front surface **2a** is less than a predetermined value, the display apparatus **2** may maintain the reflectance of the display region **2a2** at **RR1**. In a case where the illuminance near the front surface **2a** is equal to or higher than the predetermined luminance, the display apparatus **2** may decrease the reflectance of the display region **2a2** to **RR2** ($<RR1$).

[0027] The case where the illuminance in the vicinity of the front surface **2a** is equal to or higher than the predetermined luminance refers to, for example, a case where the headlight of the following vehicle is reflected at equal to or higher than the predetermined luminance or a case where sunlight is reflected at equal to or higher than the predetermined luminance in the rear image. In such a case, the display apparatus **2** can implement the automatic anti-glare by the VRM in the mirror mode. In addition, the display apparatus **2** may lower the reflectance of the display region **2a2** to **RR3** ($<<RR2$) in the display mode. As a result, in the display mode, when an image of an electro-optical member **27** (see FIG. 3) is transmitted through the front surface **2a** and displayed, the display apparatus **2** can allow the user to easily view the image.

[0028] On the front surface **2a**, the frame region **2a1** has substantially a width **W1** at a portion extending in the Y direction, and substantially a width **W2** at a portion extending in the X direction. The width **W1** and the width **W2** may be equal or different.

[0029] In the display apparatus **2**, as illustrated in FIG. 3, a transmissive member **26** is provided. The light transmittance of the transmissive member **26** is set to 5% or more. With this configuration, the width of the contour existing at the boundary between the frame region **2a1** and the display region **2a2** on the front surface **2a** can be reduced.

[0030] FIG. 3 is an XZ cross-sectional view illustrating an example of a configuration of the display apparatus **2** according to the embodiment. FIG. 3 illustrates an XZ cross section taken along line A-A in FIG. 2. In FIG. 3, the structure of the XZ cross section corresponding to the portion extending in the Y direction of the frame region **2a1** is illustrated. However, the concept of the present embodiment is similarly applicable to the structure of the YZ cross section corresponding to the portion extending in the X direction of the frame region **2a1**.

[0031] In the example illustrated in FIG. 3, the display apparatus **2** includes an optical member **21**, an electro-optical member **22**, a housing **23**, an adhesive layer **24**, a half mirror film **25**, a transmissive member **26**, an electro-optical member **27**, an adhesive layer **28**, a frame member **29**, a frame member **30**, an optical member **31**, and a case **32**. The above-mentioned VRM function is mainly implemented by the electro-optical member **22** and the half mirror film **25**.

[0032] The optical member **21** extends in a plate shape in the XY directions and has a substantially rectangular shape with the X direction as a longitudinal direction. The optical member **21** has light-transmitting properties (namely, translucency). The optical member **21** may be formed of inorganic glass containing SiO₂, or may be formed of organic glass containing a transparent resin such as polymethyl methacrylate resin (PMMA) or polycarbonate (PC). The optical member **21** is also called a cover panel, and can protect the display apparatus **2** from an external impact or the like.

[0033] The optical member **21** has the front surface **2a** on the +Z side and a back surface **21a** on

the -Z side. The front surface **2a** has the frame region **2a1** and the display region **2a2**. The frame region **2a1** is an edge portion of the front surface **2a**, and extends in a stripe shape in the Y direction or the X direction along an outer contour of the front surface **2a** (see FIG. 2). The display region **2a2** flatly extends in the XY directions inside the frame region **2a1**.

[0034] The back surface **21a** has a first region **21a1** and a second region **21a2**. The second region **21a2** is located outside the first region **21a1** in the XY directions. The first region **21a1** corresponds to the display region **2a2** and overlaps with the display region **2a2** when viewed from the Z direction. The second region **21a2** corresponds to the frame region **2a1** and overlaps with the frame region **2a1** when viewed from the Z direction.

[0035] The transmissive member **26** is a transmissive member that is provided on a peripheral edge of the back surface **21a** of the optical member **21**, which lies behind the front surface **2a** corresponding to the observation side. The transmissive member **26** has a transmittance of 5% or more. In one example, the transmissive member **26** is printed or bonded to the back surface **21a** of the optical member **21**. In one example, the transmissive member **26** is provided in the second region **21a2**. Additionally, the transmissive member **26** is formed by sequentially laminating an insulating film layer and a metal film layer on the back surface **21a** of the optical member **21**. The insulating film layer covers the end portion of the metal film layer. The insulating film layer and the metal film layer may be formed by vapor deposition. The layer structure of the transmissive member **26** will be described with reference to FIG. 4. FIG. 4 is a schematic diagram illustrating an example of the transmissive member **26** according to the embodiment.

[0036] In the present embodiment, the transmissive member **26** is constituted by an insulating film layer and a metal film layer. The insulating film layer is, for example, a silicon dioxide film layer **261** formed of silicon dioxide. The metal film layer is, for example, a chromium film layer **262** formed of chromium. The silicon dioxide film layer **261** covers the end portion of the chromium film layer **262** and contacts the end portion of the optical member **21**. The chromium film layer **262** is provided between the optical member **21** and the silicon dioxide film layer **261**.

[0037] The film thickness of the silicon dioxide film layer **261** in the present embodiment is configured to be in a range of 130 nm to 600 nm, inclusive. The film thickness of the silicon dioxide film layer **261** can be set based on the electrostatic protection performance of the display apparatus **2** and the degree of influence on the appearance hue.

[0038] When the film thickness of the silicon dioxide film layer **261** is 130 nm or more, the silicon dioxide film layer **261** covers the side surface of the metal film such as the chromium film layer **262**. With this configuration, it is possible to sufficiently suppress electricity from flowing through the metal film when the display apparatus **2** receives static electricity.

[0039] Therefore, it is possible to reduce the possibility that the electricity flowing in the metal film flows to the electro-optical member **22** and the electro-optical member **27** and thereby a trouble is caused in the electro-optical member **22** and the electro-optical member **27**. As a result, the withstand voltage of the display apparatus **2** can be improved.

[0040] In addition, by covering the side surface of the metal film such as the chromium film layer **262** with the silicon dioxide film layer **261**, it is possible to sufficiently suppress electricity from flowing through the metal film when the display apparatus **2** receives static electricity.

[0041] Moreover, in the display apparatus **2**, by setting the film thickness of the silicon dioxide film layer **261** to 600 nm or less, it is possible to suppress the influence of the hue of the silicon dioxide film layer **261** on the appearance of the image or the mirror image displayed on the display apparatus **2**.

[0042] Next, the light transmittance of the transmissive member **26** will be described with reference to FIG. 5. FIG. 5 is a table for describing the film thickness and transmittance of the transmissive member **26** according to the embodiment. FIG. 5 is a table T1 illustrating the thickness and transmittance of the chromium film layer **262** in a case where the transmissive member **26** according to the embodiment is constituted by the chromium film layer **262**.

[0043] As illustrated in FIG. 5, in a case where the film thickness of the chromium film layer **262** is set to 20 nm, the light transmittance is 15.6%. In a case where the film thickness of the chromium film layer **262** is set to 25 nm, the light transmittance is 11.1%. In a case where the film thickness of the chromium film layer **262** is set to 30 nm, the light transmittance is 8.0%. In a case where the film thickness of the chromium film layer **262** is set to 35 nm, the light transmittance is 5.8%.

[0044] Therefore, by setting the film thickness of the chromium film layer **262** to 35 nm or less, the transmissive member **26** having a transmittance of 5% or more can be realized. The film thickness of the chromium film layer **262** may be set to 20 nm or more. With this configuration, the transmittance of the transmissive member **26** can be 16% or less, and a member provided behind the transmissive member **26** can be made difficult to be visually recognized by the user.

[0045] Returning to FIG. 3, the transmissive member **26** is disposed between the optical member **21** and the electro-optical member **22** in the Z direction. The transmissive member **26** is disposed between the second region **21a2** and the electro-optical member **22** in the Z direction. The end portion of the transmissive member **26** is disposed between the end portion of the optical member **21** and a protruding portion **231** of the housing **23** described later.

[0046] The transmissive member **26** has a substantially rectangular outer contour whose longitudinal direction is the X direction. The transmissive member **26** has an opening corresponding to the display region **2a2**. The transmissive member **26** contacts the back surface **21a** of the optical member **21** on the +Z side and covers the second region **21a2**. The transmissive member **26** covers the +Z-side surface of the electro-optical member **22**.

[0047] The electro-optical member **22** is provided on the back surface side (−Z side) of the optical member **21**. The electro-optical member **22** is provided on the back surface side (−Z side) of the transmissive member **26**. The optical characteristics of the electro-optical member **22** can vary with the applied voltage. The electro-optical member **22** has a liquid crystal region **221** and a peripheral region **222**. A boundary between the liquid crystal region **221** and the peripheral region **222** in the electro-optical member **22** defines a boundary between the display region **2a2** and the frame region **2a1** on the front surface **2a**. The peripheral region **222** may be bonded to the back surface of the transmissive member **26** with a transparent adhesive such as optically clear adhesive (OCA).

[0048] The electro-optical member **22** has a front surface **22b** (an example of the first surface) on the +Z side and a back surface **22a** (an example of the second surface) on the −Z side. The front surface **22b** has a region **22b1** and a region **22b2**. The region **22b2** is located outside the region **22b1** in the XY directions. The region **22b1** corresponds to the display region **2a2** and overlaps with the display region **2a2** when viewed from the Z direction. The region **22b2** corresponds to the frame region **2a1** and overlaps with the frame region **2a1** when viewed from the Z direction.

[0049] The back surface **22a** has a region **22a1** and a region **22a2**. The region **22a2** is located outside the region **22a1** in the XY directions. The region **22a2** corresponds to the frame region **2a1** and overlaps with the frame region **2a1** when viewed from the Z direction. The region **22a1** corresponds to the display region **2a2** and overlaps with the display region **2a2** when viewed from the Z direction.

[0050] The transmissive member **26** covers the region **22b2**. The region **22b2** corresponds to the peripheral region **222** of the electro-optical member **22**. By setting the transmittance of the transmissive member **26** to a predetermined value or less, the transmissive member **26** can shield elements, wirings, terminals, and the like disposed in the peripheral region **222** of the electro-optical member **22**. As a result, elements, wiring, terminals, and the like disposed in the peripheral region **222** of the electro-optical member **22** are less likely to be visually recognized by the user, so that the appearance of the display apparatus **2** can be improved.

[0051] A polarizing plate **41**, an adhesive member **42**, and a PET film **43** formed of a resin material are sequentially laminated on the front surface **22b** of the electro-optical member **22**, which faces the optical member **21**.

[0052] The polarizing plate **41** extends in the XY directions. The polarizing plate **41** is provided on

the back surface side ($-Z$ side) of the optical member **21**. The polarizing plate **41** is provided on the back surface side ($-Z$ side) of the adhesive member **42**. The polarizing plate **41** is provided between the electro-optical member **22** and the PET film **43**. The polarizing plate **41** is bonded to the electro-optical member **22** by an adhesive layer (not illustrated). The adhesive layer includes, for example, a transparent adhesive such as OCA.

[0053] The adhesive member **42** extends in the XY directions. The adhesive member **42** is provided on the back surface side ($-Z$ side) of the optical member **21**. The adhesive member **42** is provided on the back surface side ($-Z$ side) of the PET film **43**. The adhesive member **42** serves to bond the polarizing plate **41** and the PET film **43** together.

[0054] The PET film **43** is a sheet member formed of a resin material containing PolyEthylene Terephthalate. The PET film **43** is an example of the retardation film in the present disclosure. The PET film **43** has a retardation value that is in a range of 2000 nm to 4000 nm, inclusive. The retardation value refers to a value representing the magnitude of the phase difference between the polarized component along the fast axis and the polarized component along the slow axis of the PET film **43**.

[0055] For instance, a retardation value of 2000 nm represents that a phase difference of 2000 nm exists between the polarized component along the fast axis and the polarized component along the slow axis. The PET film **43** is, for example, 100 μm .

[0056] The PET film **43** extends in the XY directions. The PET film **43** is provided on the back surface side ($-Z$ side) of the optical member **21**. The PET film **43** is bonded to the back surface **21a** of the optical member **21** and the transmissive member **26** by an adhesive layer (not illustrated). The adhesive layer includes, for example, a transparent adhesive such as OCA.

[0057] The liquid crystal region **221** corresponds to the display region **2a2** and extends in the XY directions. The liquid crystal region **221** is, for example, a TN system liquid crystal panel. The liquid crystal panel has a configuration in which a TN type liquid crystal material is sealed between a pair of transparent substrates **44** being spaced from each other in the Z direction and extending in the XY directions.

[0058] The peripheral region **222** corresponds to the frame region **2a1** and extends so as to surround the periphery of the liquid crystal region **221** in the XY plane view. In the peripheral region **222**, the end portions of the pair of transparent substrates **44** are disposed, and a member for sealing the liquid crystal material is disposed. Of the pair of transparent substrates **44**, the $+Z$ -side surface of the end portion of the $+Z$ -side transparent substrate **44** may be bonded to the back surface of the transmissive member **26** with a transparent adhesive such as OCA. The $-Z$ -side surface of the end portion of the $-Z$ -side transparent substrate **44** may be bonded to the housing **23** with a transparent adhesive such as OCA. A transparent electrode formed of a transparent conductive material such as indium tin oxide (ITO) can be disposed on each of the pair of transparent substrates **44**.

[0059] Additionally, a control circuit or the like connected to the pair of transparent electrodes may be provided in the peripheral region **222**. The control circuit can change the alignment state of the liquid crystal molecules by changing the voltage applied to the pair of transparent electrodes, thereby changing the optical characteristics (for example, transmittance and reflectance) of the electro-optical member **22**. The VRM function is mainly implemented by change in optical characteristics of the electro-optical member **22** and an operation of light reflection and transmission by the half mirror film **25** according to the change.

[0060] The housing **23** is provided on the back surface side ($-Z$ side) of the optical member **21** and is provided outside the electro-optical member **22** in the XY directions. The housing **23** has the protruding portion **231** and a side wall portion **232**. The protruding portion **231** protrudes from the side wall portion **232** in a direction along the back surface **22a** of the electro-optical member **22**. The protruding portion **231** extends along the back surface **22a** of the electro-optical member **22**. The protruding portion **231** covers the end portion of the electro-optical member **27** and protrudes

toward the back surface of the optical member **21**. The side wall portion **232** covers an XY direction end surface **22c** of the electro-optical member **22** from the outside in the XY directions on the outside in the XY directions of the electro-optical member **22**. The side wall portion **232** extends in the YZ directions or the ZX directions (see FIG. 2) and forms an outer surface **23a** of the housing **23**. The housing **23** can be formed of a material having light shielding properties.

[0061] The adhesive layer **24** is provided between the back surface **22a** of the electro-optical member **22** and the protruding portion **231**. The adhesive layer **24** may be a member in which an adhesive is applied to a +Z-side surface and a -Z-side surface of a double-sided tape or the like. The adhesive layer **24** is provided between the region **22a2** and the protruding portion **231** to bond the peripheral region **222** of the electro-optical member **22** to the protruding portion **231**.

[0062] Thus, the optical member **21** is bonded to the electro-optical member **22** via a transparent adhesive such as OCA, the PET film **43**, etc. The adhesive layer **24** bonds the back surface **22a** of the electro-optical member **22** to the housing **23**. As a result, the optical member **21** is fixed to the housing **23** via the electro-optical member **22**. Therefore, in the display apparatus **2**, the width of the adhesive layer **24** in the X direction can be secured according to the width of the peripheral region **222** in the X direction, and the bonding area of the adhesive layer **24** can be secured wide, so that the bonding strength (for example, vibration resistance robustness) can be easily secured.

[0063] The adhesive layer **24** is provided between the back surface **22a** of the electro-optical member **22** and the protruding portion **231** to fix the optical member **21** to the housing **23** via the electro-optical member **22**. With this configuration, the width in the X direction of the adhesive layer **24** can be secured according to the width in the X direction of the peripheral region **222** while an X-direction end portion **22ci** of the electro-optical member **22** is positioned on the more outer side in the X direction. In short, the X-direction end portion **22ci** of the electro-optical member **22** can be brought close to the outer surface **23a** of the housing **23**. Accordingly, the boundary between the liquid crystal region **221** and the peripheral region **222** in the electro-optical member **22** can be located further outside in the X direction.

[0064] The half mirror film **25** is provided between the electro-optical member **22** and the electro-optical member **27** in the Z direction. The transmissive member **26** is disposed between the region **22a1** and the electro-optical member **27** in the Z direction. The half mirror film **25** has a substantially rectangular shape whose longitudinal direction is the X direction corresponding to the display region **2a2** (see FIG. 2). The half mirror film **25** contacts the back surface **22a** of the electro-optical member **22** on the +Z side and covers the region **22a1**. The half mirror film **25** may be bonded to the back surface **22a** of the electro-optical member **22** with a transparent adhesive such as OCA. The half mirror film **25** is separated from the electro-optical member **27** on the -Z side. The half mirror film **25** may be a polarization reflection layer.

[0065] In a case where the display apparatus **2** is observed from the +Z side in the mirror mode, the frame region **2a1** looks similar in color to the transmissive member **26**, and the display region **2a2** looks similar in color to the half mirror film **25**. At this time, as indicated by a dotted line in FIG. 2, a black line may be visually recognized at the boundary between the frame region **2a1** and the display region **2a2**.

[0066] The black line is considered to occur as follows. Part of the light incident from the +Z side of the front surface **2a** is reflected by the surface of the half mirror film **25** and is directed to the back surface of the transmissive member **26**. It is considered that a black line is generated by the light being shielded by the transmissive member **26** and visually recognized as a shadow of the transmissive member **26**. The width of the visually recognized black line may be about twice the width of the region where the reflected light of the half mirror film **25** may be shielded by the transmissive member **26**. In other words, when the region where the reflected light of the half mirror film **25** is shielded by the transmissive member **26** increases, the black line is easily noticeable. Since the transmissive member **26** of the present embodiment has a transmittance of 5% or more, it is difficult to shield reflected light of the half mirror film **25**. As a result, the black line

width can be suppressed, so that the black line can be made less noticeable.

[0067] The frame member **29** is provided between the housing **23** and the electro-optical member **27** in the XY directions. The frame member **29** extends along the YZ directions or the ZX directions. The frame member **29** is bent inward in the XY directions such that an end portion on the +Z side covers the electro-optical member **27** from the +Z side. The frame member **29** may be formed of material, such as metal, that can be processed into a sheet-metal. The frame member **29** may be fixed to the housing **23**.

[0068] The electro-optical member **27** is disposed on the back surface **22a** side (−Z side) of the electro-optical member **22**. The electro-optical member **27** is disposed between the electro-optical member **22** and the optical member **31** in the Z direction. The electro-optical member **27** is disposed between the frame member **29** and the frame member **30** in the Z direction. The optical characteristics of the electro-optical member **27** can be changed by applying voltage. The electro-optical member **27** has a liquid crystal region **271** and a peripheral region **272**. The electro-optical member **27** may be, for example, a display panel for image display.

[0069] The liquid crystal region **271** corresponds to the display region **2a2** and extends in the XY directions. The liquid crystal region **271** includes, for example, a TFT system liquid crystal panel **271a** and a polarizing plate **271b** provided on the +Z side thereof. The liquid crystal panel **271a** has a configuration in which a TFT type liquid crystal material is sealed between a pair of transparent substrates being separated from each other in the Z direction and extending in the XY directions. The liquid crystal region **271** may further include a polarizing plate provided on the −Z side of the liquid crystal panel **271a**.

[0070] The peripheral region **272** corresponds to the frame region **2a1** and extends so as to surround the periphery of the liquid crystal region **271** in the XY plane view. In the peripheral region **272**, end portions of the pair of transparent substrates are disposed, and a member for sealing the liquid crystal material is provided. Of the pair of transparent substrates, a −Z-side surface of an end portion of the −Z-side transparent substrate may be bonded to the frame member **30** with a transparent adhesive such as OCA. A transparent electrode formed of a transparent conductive material such as ITO can be provided on each of the pair of transparent substrates.

[0071] Additionally, a control circuit or the like connected to the pair of transparent electrodes may be provided in the peripheral region **272**. The control circuit can change the alignment state of the liquid crystal molecules by changing the voltage applied to the pair of transparent electrodes, thereby changing the optical characteristics (for example, transmittance) of the electro-optical member **27**. The image display function is mainly implemented by a change in optical characteristics of the electro-optical member **27**.

[0072] The frame member **30** is provided between the frame member **29**, and the case **32** and the optical member **31** in the XY directions. The frame member **30** extends along the YZ directions or the ZX directions. The frame member **30** has a bent portion **30a** bent inward in the XY directions such that an end portion on the +Z side covers the case **32** and the optical member **31** from the +Z side. As described above, the frame member **29** may be formed of material, such as metal, that can be processed into a sheet-metal. The frame member **30** may be fixed to the housing **23** directly or via the frame member **29**.

[0073] The adhesive layer **28** is provided between the back surface of the electro-optical member **27** and the bent portion **30a** of the frame member **30**. The adhesive layer **28** may be a member in which an adhesive is applied to a +Z-side surface and a −Z-side surface of a double-sided tape or the like. The adhesive layer **28** is provided between an outer region on the back surface of the electro-optical member **27** and the bent portion **30a** of the frame member **30**, and bonds the peripheral region **272** of the electro-optical member **27** to the bent portion **30a** of the frame member **30**.

[0074] Thus, the adhesive layer **28** bonds the back surface of the electro-optical member **27** to the frame member **30**. Thus, the electro-optical member **27** is fixed to the housing **23** via the frame

member **30**. As a result, the width in the X direction of the adhesive layer **28** can be secured according to the width in the X direction of the peripheral region **272** while the X-direction end portion **22ci** of the electro-optical member **27** is located relatively outside in the X direction. As a result, it is possible to secure a wide bonding area of the adhesive layer **28** while securing a wide display region, so that it is possible to easily secure the bonding strength.

[0075] The optical member **31** is provided on the back surface side ($-Z$ side) of the electro-optical member **27**. The optical member **31** extends in a plate shape in the XY directions and has a substantially rectangular shape whose longitudinal direction corresponds to the X direction. The optical member **31** is configured to illuminate the electro-optical member **27** from the back surface side. The optical member **31** is, for example, a backlight element.

[0076] The case **32** is provided on the back surface side ($-Z$ side) of the electro-optical member **27** and is provided on the $-Z$ side of the optical member **31**. The case **32** has a box shape with the $+X$ side open, and can accommodate the optical member **31**. The case **32** may be fixed to the housing **23** directly or via the frame member **30** and/or the frame member **29**.

[0077] Next, appearance of the transmissive member **26** and the display apparatus **2** according to the present embodiment will be described. FIG. **6** is a table illustrating an example of a relationship between transmittance and appearance of the transmissive member **26** according to the embodiment. Note that the appearance is represented by determination as to whether or not the inside of the display apparatus **2** is seen through and is visually recognized by the user, and whether or not the contour of the boundary between the frame region **2a1** and the display region **2a2** is visually recognized by the user in black. In FIG. **6**, contents of evaluation on the appearance are illustrated, which are obtained for the transmittance of the transmissive member **26** being 5%, 10%, 15%, and 20%.

[0078] In the table T2 illustrated in FIG. **6**, the items of appearance are “visibility of structures”, “visibility of the electro-optical member”, and “contour of the boundary”. The item of “visibility of structures” indicates whether or not the structures inside the display apparatus **2** are seen through and visually recognized by the user in a case where the display apparatus **2** is observed from the surface. The item of “visibility of the electro-optical member” indicates whether or not the electro-optical member **22** inside the display apparatus **2** is seen through and visually recognized by the user. The item of “contour of the boundary” indicates whether or not the contour of the boundary between the frame region **2a1** and the display region **2a2** is visually recognized by the user. Here, the structure is the transparent substrate **44**. Note that the item of appearance is not limited thereto.

[0079] In addition, in FIG. **6**, a double circular symbol, a single circular symbol, and a triangular symbol are illustrated as the evaluation results. The double circular symbol represents that, as a result of the evaluation, the display apparatus **2** is in a good state even in a case where the display apparatus **2** is used in an external environment of the vehicle **5** or the display apparatus **2** is irradiated with light stronger than light irradiated in a case of normal use in the vehicle **5**, for example. Here, the good state refers to a state in which the user cannot visually recognize the structures inside the display apparatus **2**, the electro-optical member **22**, and the contour of the boundary between the frame region **2a1** and the display region **2a2**.

[0080] The single circular symbol represents that, as a result of the evaluation, the display apparatus **2** is in a good state in any environment in a case of being used in the vehicle **5**, for example. The triangular symbol represents that, as a result of the evaluation, for example, in a case of being used in the vehicle **5**, the target object can be visually recognized by the user in some environments although it is good in a normal environment. Some environments are, for example, a case where the display apparatus **2** receives the setting sun. The target object is, for example, structures inside the display apparatus **2**, the electro-optical member **22**, and the contour of the boundary between the frame region **2a1** and the display region **2a2**.

[0081] Therefore, as illustrated in FIG. **6**, by setting the transmittance of the transmissive member **26** to 5% or more, more specifically, to be in a range of 5% to 20%, inclusive, it is possible to

reduce visibility (or transparency) of the structures or the electro-optical member **22** and the contour of the boundary between the frame region **2a1** and the display region **2a2**.

[0082] The transmittance of the transmissive member **26** may be set to 10% or more. With this configuration, the contour of the boundary between the frame region **2a1** and the display region **2a2** can be made more difficult to be visually recognized by the user. Moreover, the transmittance of the transmissive member **26** may be set to 15% or less. With this configuration, the electro-optical member **22** can be made more difficult to be visually recognized by the user.

[0083] As described above, the embodiment includes the optical member **21** with translucency, the transmissive member **26**, the electro-optical member **22**, and the housing **23**. The optical member **21** includes the front surface **2a** corresponding to an observation side and the back surface **21a** lying behind the front surface **2a**. The transmissive member **26** is provided on the peripheral edge of the back surface **21a** of the optical member **21**. The transmissive member has a transmittance of 5% or more. The electro-optical member **22** is provided on a side of the back surface **21a** of the optical member **21**. The electro-optical member **22** has optical characteristics that vary with applied voltage. The housing **23** includes the protruding portion **231**. The protruding portion **231** covers the end portion of the electro-optical member **22** and protrudes toward the back surface **21a** of the optical member **21**. The end portion of the transmissive member **26** is located between the end portion of the optical member **21** and the protruding portion **231**. Consequently, the contour of the boundary between the frame region **2a1** and the display region **2a2** can be reduced.

[0084] Note that, in the description of the present disclosure, components having the same or substantially the same functions as those described above with respect to the previously described drawings are denoted by the same reference signs, and the description thereof may be appropriately omitted. In addition, even in the case of representing the same or substantially the same portion, the dimensions and ratios may be represented differently from each other depending on the drawings. Furthermore, for example, from the viewpoint of ensuring the visibility of the drawings, in the description of each drawing, only main components are denoted by reference signs, and even components having the same or substantially the same functions as those described above in the previous drawings may not be denoted by reference signs.

First Modification

[0085] In the above-described embodiment, the end portion of the transmissive member **26** is provided between the end portion of the optical member **21** and the protruding portion **231**. However, the present invention is not limited thereto. For example, the end portion of the PET film **43** may be disposed between the end portion of the transmissive member **26** and the protruding portion **231**. FIG. 7 is a cross-sectional view illustrating an example of a configuration of a display apparatus **2** according to a first modification.

[0086] Comparing FIG. 7 with FIG. 3, the adhesive member **42** and the PET film **43** illustrated in FIG. 7 extend in the negative direction of the X axis, and the end portion of the PET film **43** is disposed between the end portion of the transmissive member **26** and the protruding portion **231**. With this configuration illustrated in FIG. 7, the contour of the boundary between the frame region **2a1** and the display region **2a2** can be reduced.

Second Modification

[0087] In the first modification described above, the end portion of the PET film **43** is disposed between the end portion of the transmissive member **26** and the protruding portion **231**. However, the present invention is not limited thereto. For example, the end portion of the polarizing plate **41** may be disposed between the end portion of the PET film **43** and the protruding portion **231**. FIG. 8 is a cross-sectional view illustrating an example of a configuration of a display apparatus **2** according to a second modification.

[0088] Comparing FIG. 8 with FIG. 7, the polarizing plate **41** in FIG. 8 extends in the negative direction of the X axis, and the end portion of the polarizing plate **41** is disposed between the end portion of the PET film **43** and the protruding portion **231**. With this configuration illustrated in

FIG. 8, the contour of the boundary between the frame region 2a1 and the display region 2a2 can be reduced.

[0089] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A display apparatus comprising: an optical member with translucency, the optical member including a front surface corresponding to an observation side and a back surface lying behind the front surface; a transmissive member provided on a peripheral edge of the back surface of the optical member, the transmissive member having a transmittance of 5% or more; an electro-optical member provided on a side of the back surface of the optical member, the electro-optical member having optical characteristics varying with applied voltage; and a housing including a protruding portion, the protruding portion covering an end portion of the electro-optical member and protruding toward the back surface of the optical member, wherein an end portion of the transmissive member is located between an end portion of the optical member and the protruding portion.
2. The display apparatus according to claim 1, wherein the electro-optical member includes a first surface facing the optical member and a second surface lying behind the first surface, a polarizing plate is provided on the first surface of the electro-optical member, and a retardation film with a resin material is provided on the polarizing plate.
3. The display apparatus according to claim 2, wherein an end portion of the retardation film is located between the end portion of the transmissive member and the protruding portion.
4. The display apparatus according to claim 3, wherein an end portion of the polarizing plate is located between the end portion of the retardation film and the protruding portion.
5. The display apparatus according to claim 1, wherein the transmissive member includes an insulating film layer and a metal film layer that are laminated on the back surface of the optical member, and the insulating film layer covers an end portion of the metal film layer.
6. The display apparatus according to claim 2, wherein the transmissive member includes an insulating film layer and a metal film layer that are laminated on the back surface of the optical member, and the insulating film layer covers an end portion of the metal film layer.
7. The display apparatus according to claim 3, wherein the transmissive member includes an insulating film layer and a metal film layer that are laminated on the back surface of the optical member, and the insulating film layer covers an end portion of the metal film layer.
8. The display apparatus according to claim 4, wherein the transmissive member includes an insulating film layer and a metal film layer that are laminated on the back surface of the optical member, and the insulating film layer covers an end portion of the metal film layer.
9. The display apparatus according to claim 5, wherein a film thickness of the metal film layer is in a range of 20 nm to 35 nm, inclusive, and a film thickness of the insulating film layer is in a range of 130 nm to 600 nm, inclusive.
10. The display apparatus according to claim 6, wherein a film thickness of the metal film layer is in a range of 20 nm to 35 nm, inclusive, and a film thickness of the insulating film layer is in a range of 130 nm to 600 nm, inclusive.
11. The display apparatus according to claim 7, wherein a film thickness of the metal film layer is in a range of 20 nm to 35 nm, inclusive, and a film thickness of the insulating film layer is in a

range of 130 nm to 600 nm, inclusive.

12. The display apparatus according to claim 8, wherein a film thickness of the metal film layer is in a range of 20 nm to 35 nm, inclusive, and a film thickness of the insulating film layer is in a range of 130 nm to 600 nm, inclusive.

13. The display apparatus according to claim 1, wherein the transmittance of the transmissive member is in a range of 5% to 20%, inclusive.

14. The display apparatus according to claim 2, wherein the transmittance of the transmissive member is in a range of 5% to 20%, inclusive.

15. The display apparatus according to claim 3, wherein the transmittance of the transmissive member is in a range of 5% to 20%, inclusive.

16. The display apparatus according to claim 4, wherein the transmittance of the transmissive member is in a range of 5% to 20%, inclusive.

17. The display apparatus according to claim 5, wherein the transmittance of the transmissive member is in a range of 5% to 20%, inclusive.
